

DocuFlow AI — Document Intelligence Test File

Synthetic test document for PDF parsing, OCR, layout, table extraction, chunking, embeddings and semantic search.

1. Project Overview

DocuFlow AI is a document intelligence platform that converts unstructured documents into structured, searchable information. The ingestion pipeline accepts documents, extracts their content and structure, creates meaningful chunks, generates embeddings, and stores the resulting vectors for semantic search.

Primary objective: make complex documents usable by downstream AI and RAG systems without losing headings, pages, tables, figures, and source information.

2. Processing Pipeline

Stage	Input	Output
1. Upload	PDF / DOCX / image	Validated document
2. Parsing	Original document	Structured document
3. OCR	Scanned pages	Recognized text
4. Normalization	Raw extracted content	Clean content
5. Chunking	Structured content	Semantic chunks
6. Embeddings	Text chunks	Vector representations
7. Indexing	Vectors + metadata	Searchable records

3. System Components

The application uses a Next.js frontend and a FastAPI backend. PostgreSQL stores application metadata, Redis and Celery handle asynchronous processing, MinIO provides S3-compatible object storage, and Qdrant stores vector embeddings for semantic retrieval.

4. Sample Dataset

This synthetic dataset tests table extraction, numeric values, categories, statuses, and metadata preservation.

Document	Category	Pages	Chunks	Status
AI_Research_2026.pdf	Research	24	186	Completed
Cloud_Architecture.pdf	Cloud	18	142	Completed
Security_Guidelines.pdf	Security	31	241	Processing
ML_Lab_Manual.pdf	Education	12	97	Completed
Annual_Report.pdf	Business	46	318	Queued

5. Example Content for Semantic Search

Document Intelligence

Document intelligence combines document parsing, OCR, layout understanding, information extraction, and search technologies to convert unstructured files into machine-readable knowledge.

Retrieval-Augmented Generation

A retrieval-augmented generation system retrieves relevant information from a knowledge base and provides that information to a language model as context. High-quality ingestion and chunking are important for retrieval accuracy.

Vector Search

Vector search represents text as numerical embeddings. Queries and document chunks can be compared using vector similarity, allowing semantically related content to be retrieved even when exact keywords do not match.

Production Considerations

A production ingestion platform should handle retries, large files, duplicate documents, authentication, authorization, observability, failures, storage lifecycle management, and data isolation.

6. Test Questions

1. What technologies are used for asynchronous document processing?
2. Which database stores application metadata?
3. Which system stores vector embeddings?
4. How many chunks are reported for AI_Research_2026.pdf?
5. Why is semantic chunking important for RAG?
6. What should happen if the same document version is processed twice?

END OF TEST DOCUMENT